T06 Section 4 Analytical Methodology

Predictive Maintenance and Machine Failure Analysis

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4. Analytical Methodology

4.1 Analytical Objective

The project asks whether available machine operating measurements can identify machine-failure records better than a simple baseline. The binary outcome is Machine failure. Permitted predictors are Type, Air temperature [K], Process temperature [K], Rotational speed [rpm], Torque [Nm], and Tool wear [min]. UDI and Product ID are excluded as identifiers. TWF, HDF, PWF, OSF, and RNF are excluded because they are failure-mode outcome fields and could disclose the target.

4.2 Data Preparation and Quality Rule

The submitted AI4I file contains 10,000 records. The current cleaning rule checks whether Machine failure agrees with whether any of the five failure-mode flags are active. This identifies 27 inconsistent records and leaves 9,973 observations. The retained data contain no missing values and no duplicate rows. The exclusion is a documented methodological choice, not proof the 27 records are objectively invalid; later sensitivity analysis should evaluate its effect.

4.3 Train/Test Design and Preprocessing

The cleaned dataset uses an 80/20 stratified train/test split with random_state=42, producing 7,978 training records and 1,995 holdout records. The holdout contains 66 failures and 1,929 non-failures. Numeric predictors are standardized using training-set parameters and Type is one-hot encoded inside the pipeline, preventing the holdout set from fitting preprocessing parameters.

4.4 Baseline and First Classification Model

The baseline is a majority-class classifier that always predicts No Failure. It provides a transparent comparison and demonstrates why accuracy alone can mislead with an imbalanced outcome. The first classifier is Logistic Regression with balanced class weights. It is an interpretable binary-classification starting point and is treated as a benchmark, not a final or optimal model.

4.5 Evaluation Measures and Decision Rule

Evaluation uses accuracy, failure-class precision, recall, F1, false negatives, false positives, and the confusion matrix. Recall measures actual failures detected; precision measures how many predicted failures are actual failures. False negatives are missed failures and false positives are false alarms. For this first benchmark, improvement requires failure detection beyond the majority baseline, particularly nonzero recall and F1, while false alarms must remain visible in the decision evidence.

4.6 Baseline and Model Results

Model	Accuracy	Precision	Recall	F1	Missed	False Alarms
No-Failure Baseline	96.69%	0.00%	0.00%	0.000	66	0
Balanced Logistic Regression	81.40%	13.60%	86.36%	0.235	9	362

The baseline achieves 96.69% accuracy but detects 0 of 66 failures. Logistic Regression detects 57 of 66 failures (86.36% recall), misses 9, and produces 362 false alarms with 13.60% precision. This supports only the conclusion that failure detection improved over the baseline; it does not establish an operationally acceptable false-alarm rate.

4.7 Reproducibility and Interpretation

The following validation record documents the technical corrections and model evaluations completed during Week 9. The team used the professor's feedback from T06 to identify areas requiring improvement and reviewed the updated modeling results to determine whether the changes addressed those concerns.

Architecture and Technical Artifact



Figure 1. Smallest complete Week 6 analytical pipeline.

Text alternative: AI4I records are validated and cleaned; six permitted predictors and Machine failure are selected; a stratified 80/20 split is created; a no-failure baseline and balanced Logistic Regression classifier are evaluated; metrics and validation evidence are produced.

Setup, Sample Input, and Expected Output

1. Keep ai4i2020.csv, t06_pipeline.py, and requirements.txt together.
2. Create or activate a Python environment.
3. Run: pip install -r requirements.txt
4. Run: python t06_pipeline.py
5. Review the labeled metrics and confusion-matrix counts.

sample_input.csv contains ten public records limited to modeling fields. Expected output is a validated cleaned dataset, six permitted predictors, a reproducible split, baseline metrics, classifier metrics, and confusion-matrix counts.

Validation Record

Validation Question	Expected Result	Actual Result	Status	Evidence
Does the leakage check examine the actual model predictors?	The check should inspect X.columns and reject prohibited failure indicators.	The updated notebooks check X.columns against the target and failure-mode indicators.	Pass	Leakage-check update
Does removing inconsistent records affect model performance?	Compare results from the original 10,000 records and the filtered 9,973 records.	Random Forest recall increased from 77.94% to 78.79%, and its false-positive rate decreased from 1.92% to 1.45%.	Pass	Dataset comparison update
Does Random Forest meet the proposed recall requirement?	Recall must be at least 75%.	The filtered Random Forest model achieved 78.79% recall.	Pass	Modeling notebooks
Does Random Forest meet the proposed false-positive-rate limit?	False-positive rate must not exceed 2%.	The filtered Random Forest model achieved a 1.45% false-positive rate.	Pass	Modeling notebooks
Can the full workflow be reproduced from a fresh environment?	All required files and dependencies should allow successful execution.	The updated notebooks contain saved outputs, but independent reproduction has not been confirmed.	Pending	Project repository

Limitations Note

The evidence supports a reproducible pipeline and improved failure detection on this holdout split. It does not establish real-world maintenance performance, causality, an optimal model or threshold, or an acceptable false-alarm rate. AI4I is synthetic. The 27-record exclusion should be tested as a sensitivity decision. The first classifier has low precision and many false positives, and no real stakeholder cost ratio has been established for missed failures versus false alarms.

Current Work Plan and Contribution Record

Owner	Task	Status	Dependency	Acceptance Criterion	Contribution Evidence
Keeshaun Cobb	Section 4 integration, validation, limitations, reproducibility evidence, final submission.	Completed for T06	A04 design + AI4I data	Package connects method, executed results, limitations, evidence.	T06 report and final integration
Brett Harrison	Prior data preparation/first-build verification; next refine classifiers.	Prior foundation complete	Clean predictor/outcome definition	Future models use permitted predictors and	A04 vertical-slice work

Owner	Task	Status	Dependency	Acceptance Criterion	Contribution Evidence
				comparable metrics.	
Wynn Nguyen	Prior questions/hypotheses; next interpret model evidence.	Prior design complete	Validated model output	Interpretation addresses questions without overstating conclusions.	A04 research design
Zach Lee	Prior evidence/visual planning; next accessible result visuals.	Planning complete	Validated metrics	Visuals labeled, captioned, with text alternatives.	A04 evidence/visual planning

Sources and AI-Use Disclosure

Dataset: UCI Machine Learning Repository, AI4I 2020 Predictive Maintenance Dataset. The course identifies AI4I as synthetic, so conclusions are limited accordingly. Project evidence also includes the team's earlier A04 cleaning and predictor-selection work.

AI use: ChatGPT was used to help organize Section 4, draft and execute the reproducible Python pipeline, create the architecture diagram, structure validation evidence, and revise the report. Numerical claims were verified by executing the pipeline against the submitted AI4I CSV. Material revisions included replacing hypothetical results with executed metrics, labeling the 27-record exclusion as a methodological choice, and limiting conclusions to the observed synthetic-data evidence.